

PROFESSIONAL SUMMARY

Skilled and hardworking, AI/ML Engineer with a Masters in Data Analytics passionate about working on GenAI models mainly in text and images, Has done Significant work on Large Language models, specialising in Indic Languages. Helped to reduce the tokenization cost of the Indic Languages found simpler ways for optimization of the foundational model and specialized in building any RAG solutions. I know my way around data from scraping and feature selection to analysing and visualizing the data along with machine learning techniques.

EDUCATION

- *Master of Science in Data Analytics, July 2022 - May 2024*
Christ University, Bengaluru.
- *B.Sc in Computer science, June 2019 - June 2022*
SRM Arts and Science College, Chennai.

SKILLS

- **Programming Languages:** Python, R
- **Database Management:** SQL, PostgreSQL, Hadoop and spark
- **Vector Databases:** Deeplake, Weaviate, Milvus, ChromaDB
- **Frameworks:** Langchain, Tensorflow, Keras, PyTorch
- **Statistical and probability Techniques:** Hypothesis Testing (parametric and non-parametric), Correlation Analysis, Bayes Theorem, Probability Distributions and Conditional Probability
- **Machine Learning:** NLP, Tokenization, Regression, KNN, XGBoost
- **Deep Learning:** Neural Networks, CNN, RNN, GAN, LSTM.
- **Version Control:** Git
- **Software Proficiency:** MS Office suite - Word, Advanced Excel, Power point and Outlook
- **Data Visualization:** Microsoft Power BI, Tableau and Google LookerStudio

WORK EXPERIENCE

- **AI/ML Intern(Founders Office) - Vizzhy, Bangalore:** **May 2024 - Current**
 - Worked on LLM tokenization techniques specific on Indic language that reduce computation cost. And I have also worked on various optimisations of the foundational model used on the chatbot Hanooman, which is available on Play Store.
 - Have done numerous Scrapping, Pre Processing and feature selection of various kinds of data and worked with different deeplearning and NLP frameworks(Tf, Pytorch, Langchain), vector storage (Weaviate, Milvus, Deeplake) and cloud storage technologies (AWS), Pipelining vast amount data for easy retrieval, I've utilized Nvidia H100s provided by the organization either for a vector search RAG usecase or for finetuning of the model.
 - As a team we have worked on government projects and my major contribution was on Implementation of different RAG frameworks for client's use case that can read through data of different format from different databases and have built a production-grade application.
 - Contributed in the founder's office team on a few Market Research, Product strategy, Financial forecasting and analytics.

- **Data Analyst - Augie, Bangalore:**

Sep 2022 - Nov 2022

- Have worked majorly on forcasting, preprocessing and visulization of E-commerce data of the company.
- Conducted a comprehensive analysis of customer demographics and purchasing behaviour, resulting in a targeted marketing campaign that increased sales
- Developed and Implemented data quality checks that reduced errors in reports

PROJECTS

- **Model Combat (<https://github.com/wherearethegiggles/Model-Combat>):**
 - The Model Combat is a different approach on optimization of foundational model. Proving that Direct preference optimization can be more efficient than RLHF
 - The application built on Gradio inspired from LYMSys that compares responses from different AI models (GPT4o, Claude and Cohere) from a set of questions on specific use cases. Users can vote on whichever model gives a better response or remark on the responses. The models are placed on a leaderboard based on the votes, the selected responses are stored as a dataset.
 - The user preferred responses can be used to frame a instruct-dataset that can be used for model optimization on specific use cases.
- **Phi-3 RAG (<https://github.com/wherearethegiggles/Phi-3-RAG->):**
 - A RAG application that can take any number of pdfs on storage and act as a chatbot that provides accurate answers for questions on those documents uploaded.
 - I have used Phi-3 as the LLM for it's adaptability, done the whole orchestration using Langchain framework and I've used nearest vector search metod with deeplake and used the LLM for streaming the output text. With adding more detailing on the orchestration and techniques I've achieved good results without the need to finetune the model.